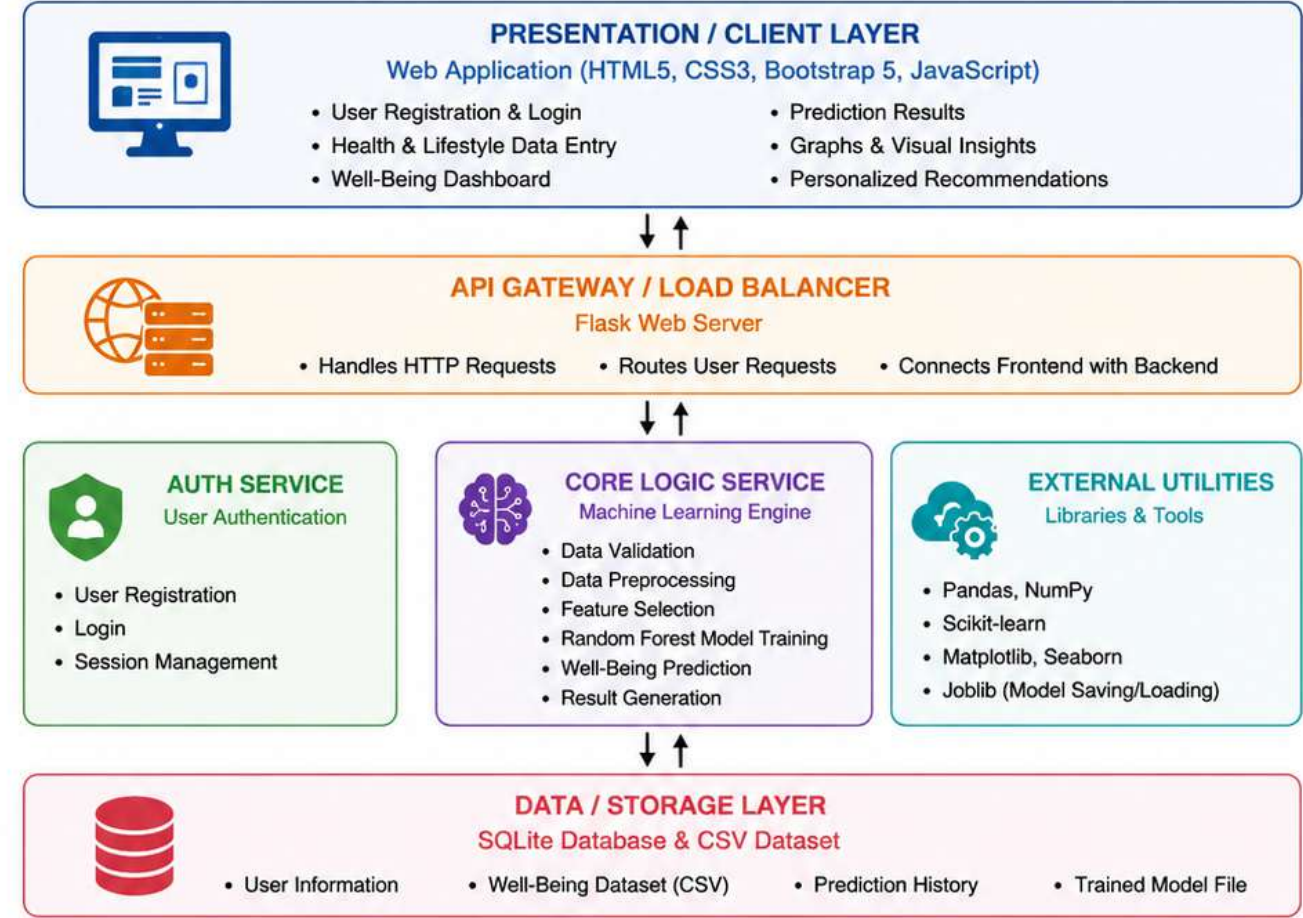


Solution Architecture

Date	15 March 2026
Team ID	xxxxxx
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Solution Architecture Diagram:



Component Description Table:

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface for registration, data input, prediction results, dashboard, graphs, and personalized recommendations.	HTML5, CSS3, Bootstrap 5, JavaScript
API Gateway	Receives HTTP requests from the client, routes them to the appropriate backend functions, and returns prediction results.	Flask
Core Logic Service	Handles authentication, data validation, preprocessing, feature selection, trains Random Forest model, predicts well-being score, and generates results.	Python, Pandas, NumPy, Scikit-learn, Joblib
Database / Storage Layer	Stores user details, well-being dataset, prediction history, and the trained machine learning model.	CSV Dataset, SQLite, Joblib (Model File)